## MC 501 Inferencing Data in R rubric

**100 points.** Due Monday, December 7, 11:59 PM.

**How this is scored.** Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: 700 to 1,100 words of your own writing (hypotheses, justifications, diagnostics, the four reports and the two paragraphs), not counting code or output. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- The worksheet is 13-inferencing-data-mc501.qmd. Its settings block gives `group2` your hand-coded variable, so one of the four tests is on the thirty messages you coded; the calls to fill are the tests, their effect sizes, and the checks behind them. The Inferencing Data risks guide shows the traps on this job and how to check for each.
- The worksheet names the four pairs of variables, one of them your hand-coded variable against message length; the variable types decide the test, and naming it with its reason is your part.
- Run all four tests: a chi-square test of independence with `chisq.test()` on an `xtabs()` table, tidied with `broom::tidy()`, and Cramér's V from `effectsize::cramers_v()`; an independent t-test with `t.test()` (Welch's version) tidied with `broom::tidy()`, and Cohen's d from `effectsize::cohens_d()`; a one-way ANOVA with `aov()` and `summary()`, `oneway.test()` beside it, then `TukeyHSD()`; and a linear regression with `lm()` and `summary()`, with `lm.beta()` for the standardized coefficient. `v2v::run_chi_square()` and `v2v::run_t_test()` print the same numbers in one labeled block and belong after the plain calls, as a check.
- The Test group of the R Companion's Write R code writes each test in this form for the settings-block names, including `oneway.test()`, both ANOVA effect sizes with their intervals, `TukeyHSD()`, and `lm.beta()`, and its Power analysis for your sample size writes the `pwr.t.test()` call behind the power statistics; the residual plots from plot(model) you write yourself. Its Check my test choice reports whether a test you have already chosen fits your variable types, and the rejected alternative and its reason are yours; its APA numbers to check your sentence prints a test's numbers in APA form so you can check the sentence you wrote, and the four reports and the two paragraphs are your own wording.

- The effectsize package returns each interval: `cramers_v()` for the chi-square, `cohens_d()` for the t-test, `omega_squared()` and `eta_squared()` on the ANOVA, and `eta_squared()` on the regression model, which for one predictor is R-squared. Check my setup in the R Companion installs it, or install.packages("effectsize") does.
- Every script opens with a settings block: load the data, rename the variables you use to outcome, `group2`, `group3`, predictor, `cat_a`, and `cat_b`, and make the categorical ones factors. The code below the block uses only those names. This is what lets the same script run on your own data for the White Paper.
- How this loses points: going wide instead of deep. Four tests each reported with their assumptions, effect size, and interval score above four tests reported as four bare sentences.
- What it feeds: your White Paper, which runs this same script on your own variables after you change its settings block. The Twitch results stay here; the paper reports only what your own data shows, and only the tests your questions call for.

*Inferencing Data in R, 100 points*

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Hypotheses**<br>8 points | **8 points.** Four hypotheses in plain English, written in Part 0 before any code, one for each test (a chi-square test of independence, an independent t-test, a one-way ANOVA across three or more groups, and a linear regression with a continuous outcome), each naming its variables and the relationship you expect. | **6 to 7 points.** Four stated before the code, but one variable is vague, or one hypothesis is a topic instead of a relationship. | **4 to 5 points.** Fewer than four, or stated after the results. | **0 to 3 points.** Missing. |
| **Test choice against a rejected alternative**<br>12 points | **12 points.** Each of the four tests justified against one alternative you rejected and why. | **9 to 11 points.** The right four tests, but a rejected alternative or its reason is missing on one. | **6 to 8 points.** A test that does not fit its hypothesis's variable types. | **0 to 5 points.** No tests chosen. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Assumptions and diagnostics**<br>20 points | **20 points.** Each test's assumptions checked with the diagnostic shown rather than asserted (expected cell counts for the chi-square, the two groups' spread for the t-test, equal variances across groups for the ANOVA, with Welch's ANOVA from `oneway.test()` where they differ, and the residual plots from plot(model) for the regression), and what you did about any that fail. | **15 to 19 points.** Checked and shown, but nothing said about one that fails. | **10 to 14 points.** Assumptions listed or asserted ("the data are normal") without being shown, or checked for some tests only. | **0 to 9 points.** Missing. |
| **APA reporting**<br>12 points | **12 points.** All four tests run in your document and are reported in APA format with the statistic, degrees of freedom, p, and n; for the ANOVA, Tukey's HSD names which groups differ. | **9 to 11 points.** One of the four pieces is missing on a test, or the Tukey step is missing. | **6 to 8 points.** The statistic and p only, or numbers pasted in instead of produced by the code. | **0 to 5 points.** Not reported. |

| Criterion | Exemplary | Proficient | Developing | Beginning |
|---|---|---|---|---|
| **Effect sizes with their intervals**<br>20 points | **20 points.** Each effect size with its 95% confidence interval, in the units of the data as well as by its conventional label: Cramér's V; Cohen's d; omega-squared with eta-squared beside it; and for the regression R-squared, the slope in the outcome's units, and the standardized coefficient. | **15 to 19 points.** All four with intervals, but one carries the label only, or eta-squared or the standardized coefficient is missing. | **10 to 14 points.** Computed without intervals, or the wrong measure for a test. | **0 to 9 points.** Missing. |
| **Power for one focal test**<br>10 points | **10 points.** For one focal test (the worksheet uses the t-test on your hand-coded variable): what effect this sample could detect, and what that means for a null result. | **8 to 9 points.** The detectable effect, without what it means for a null result. | **5 to 7 points.** A power figure with no link to your own design (a generic eighty percent). | **0 to 4 points.** Missing. |
| **Significance against size**<br>10 points | **10 points.** A paragraph that would survive a reviewer, separating statistical significance from practical size across all four results. | **8 to 9 points.** Separates the two, but for some of the results only. | **5 to 7 points.** Treats a small p as a large effect. | **0 to 4 points.** Missing. |
| **What the results do not license**<br>8 points | **8 points.** One paragraph, including the gap between the number of messages and the number of channels behind any channel-level variable. | **6 to 7 points.** A limit, but the gap between messages and channels is not named. | **4 to 5 points.** A generic limit ("more research is needed"). | **0 to 3 points.** Missing. |

### Before you submit

Inferential Check recomputes a chi-square or a t test from your own table or group figures and returns the statistic, the degrees of freedom, p, the effect size, the assumptions your data may be breaking, and the APA sentence to compare against what you wrote. It is a second opinion on your R output, not a replacement for it. It does not grade, and nothing you enter leaves your machine. It does not cover ANOVA or regression, so check those two against the worked examples in Chapter 13.